C3 Bonding – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Types of Bonding and Ions **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Ionic bonding				Electrostatic attraction between oppositely charged ions
2	Covalent bond				Shared pair of electrons
3	Dative covalent (coordinate) bond				Shared pair of electrons both from one atom
4	Metallic bonding				Attraction between positive ions and delocalised electrons
5	Example of a species with a dative bond				NH_4^+
6	Formula of the sulfate ion				SO_4^{2-}
7	Formula of the nitrate ion				NO_3^-
8	Formula of the carbonate ion				CO_3^{2-}
9	Formula of the ammonium ion				NH_4^+
10	Formula of the hydroxide ion				OH^-

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Structures **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Structure of sodium chloride				Giant ionic lattice
2	Structure of diamond				Giant covalent (macromolecular)
3	Structure of graphite				Giant covalent, in layers
4	Structure of iodine				Simple molecular
5	Structure of magnesium				Giant metallic lattice
6	Bonds per carbon atom in diamond				4
7	Bonds per carbon atom in graphite				3
8	Why graphite conducts electricity				Delocalised electrons
9	Why ionic compounds conduct when molten				Ions free to move
10	Why metals are malleable				Layers of ions slide

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Shapes of Molecules **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	2 bonding pairs, 0 lone pairs				Linear, 180°
2	3 bonding pairs, 0 lone pairs				Trigonal planar, 120°
3	4 bonding pairs, 0 lone pairs				Tetrahedral, 109.5°
4	3 bonding pairs, 1 lone pair				Trigonal pyramidal, 107°
5	2 bonding pairs, 2 lone pairs				Bent, 104.5°
6	5 bonding pairs, 0 lone pairs				Trigonal bipyramidal, 120° and 90°
7	6 bonding pairs, 0 lone pairs				Octahedral, 90°
8	4 bonding pairs, 2 lone pairs				Square planar, 90°
9	Bond angle reduction per lone pair				About 2.5°
10	Order of electron pair repulsion				LP–LP > LP–BP > BP–BP

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C3 Bonding – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 4: Electronegativity and Intermolecular Forces **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Electronegativity				Power of an atom to attract the pair of electrons in a covalent bond
2	Most electronegative element				Fluorine
3	Trend in electronegativity across a period				Increases
4	Trend in electronegativity down a group				Decreases
5	Polar bond				Unequal sharing of electrons
6	Three types of intermolecular force				Induced dipole–dipole (van der Waals), permanent dipole–dipole, hydrogen bonding
7	Atoms H must bond to for hydrogen bonding				N, O or F
8	Strongest intermolecular force				Hydrogen bonding
9	Why ice is less dense than water				Open lattice held by hydrogen bonds
10	Factor increasing van der Waals strength				More electrons

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C3 Bonding – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 5: Named Shapes and Polarity **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Shape of CH_4				Tetrahedral
2	Shape of BF_3				Trigonal planar
3	Shape of NH_4^+				Tetrahedral
4	Shape of PCl_5				Trigonal bipyramidal
5	Shape of SF_6				Octahedral
6	Shape of XeF_4				Square planar
7	Bond angle in H_2O				104.5°
8	Bond angle in NH_3				107°
9	Why CO_2 is non-polar				Symmetrical, so dipoles cancel
10	Example of a non-polar molecule with polar bonds				CCl_4 (or CO_2)

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.